

Dunyu Liu

SENIOR COMPUTATIONAL GEOSCIENTIST

Institute for Geophysics, The University of Texas at Austin

✉ dliu@ig.utexas.edu | 🌐 <https://github.com/dunyuliu> | 🔗 <https://www.linkedin.com/in/dunyu-liu/> | 🐦 @DunyuLiu

Education

Texas A&M University

PHD GEOPHYSICS

College Station, TX, USA

2014 - 2018

Peking University

MS IN GEOPHYSICS

Beijing, China

2011 - 2014

Peking University

BS IN GEOPHYSICS

Beijing, China

2007 - 2011

Professional Experience

2025-
present

Senior Computational Geoscientist, Institute for Geophysics, The University of Texas at Austin

2021-2025

Computational Geoscientist, Institute for Geophysics, The University of Texas at Austin

2018-2020

Postdoctoral researcher, Department of Geology & Geophysics, Texas A&M University

2014-2018

Research and Teaching Assistant, Department of Geology & Geophysics, Texas A&M University

2011-2014

Research and Teaching Assistant, Institute for Theoretical Geophysics, Peking University

2009-2011

Undergraduate Research and Teaching Assistant, Institute for Theoretical Geophysics, Peking University

Publications

2026

- [19] Dunnigan, A. H., **Liu, D.**, Steinbrügge, G. B., Rivoldini, A., Dumbery, M., Cao, H., Soderlund, K. M.: Interior models of Mercury and conditions for iron snow formation in a Fe-S-Si core. *J. Geophys. Res. - Planets*, 131, e2025JE009368, 2026.

2025

- [18] **Liu, D.**, Becker, T.W.: Earthquake Rupture Dynamics From Graph Neural Networks. *J. Geophys. Res. - Solid Earth*, 130(12), e2025JB031981, 2025.

- [17] White, K., **Liu, D.**, Persad, G.: Absence of Aerosol Indirect Effect Dependence on Background Climate State in NCAR CESM2. *J. Climate*, **38**, 147-163, <https://doi.org/10.1175/JCLI-D-23-0755.1>, 2025.

2024

- [16] Thirumalai, K., DiNezio, P.N., Partin, J.W., **Liu, D.**, Costa, K., and Jacobel, A.: Future increase in extreme El Niño supported by past glacial changes. *Nature*, <https://doi.org/10.1038/s41586-024-07984-y>, 2024.

- [15] **Liu, D.**, Puel, S., Becker, T.W., Moresi, L.: Analytical and numerical models of viscous anisotropy: a toolset to constrain the role of mechanical anisotropy for regional tectonics and fault loading. *Geophys. J. Int.*, 239(2), 950-963, <https://doi.org/10.1093/gji/ggae296>, 2024.

- [14] Puel, S., Becker, T.W., Villa, U., Ghattas, O., **Liu, D.**: Volcanic arc rigidity variations illuminated by coseismic deformation of the 2011 Tohoku-oki M9. *Science Adv.*, 10, eadl4264, <https://doi.org/10.1126/sciadv.adl4264>, 2024.

- [13] Puel, S., Becker, T.W., Villa, U., Ghattas, O., **Liu, D.**: An adjoint-based optimization method for jointly inverting heterogeneous material properties and fault slip from earthquake surface deformation data. *Geophys. J. Int.*, 236(2), 778-797, <https://doi.org/10.1093/gji/ggad442>, 2024.

2023

- [12] Xu, X., **Liu, D.**, Lavier, L.: Constraining fault damage zone properties from geodesy: A case study near the 2019 Ridgecrest earthquake sequence. *Geophys. Res. Lett.*, 50, e2022GL101692, 2023.

2022

- [11] Puel, S., Khattatov, E., Villa, U., **Liu, D.**, Ghattas, O., Becker TW: A mixed, unified forward/inverse framework for earthquake problems: fault implementation and coseismic slip estimate. *Geophys. J. Int.*, 230(2), 733-758, 2022.
- [10] Jiang, J. et al.: Community-driven code comparisons for three-dimensional dynamic modeling of sequences of earthquakes and aseismic slip. *J. Geophys. Res. - Solid Earth*, 127(3), e2021JB023519, 2022.
- [9] **Liu, D.**, Duan, B., Scharer, K., Yule, D.: Observation-constrained multicycle dynamic models of the southern San Andreas and the northern San Jacinto faults: addressing complexity in paleoearthquake extent and recurrence with realistic 2D fault geometry. *J. Geophys. Res. - Solid Earth*, 127(2), e2021JB023420, 2022.

2021

- [8] **Liu, D.**, Duan, B., Prush, VB, Oskin, M., Liu-Zeng, J.: Observation-constrained multicycle dynamic models of the Pingding Shan earthquake gate along the Altyn Tagh Fault. *Tectonophysics*, 814, 228948, 2021.

2020

- [7] **Liu, D.**, Duan, B., Luo, B.: EQsimu, a 3-D finite element dynamic earthquake simulator for multicycle dynamics of geometrically complex faults governed by rate- and state-dependent friction. *Geophys. J. Int.*, 220(1), 598-609, 2020.
- [6] Luo, B., Duan, B, **Liu, D.**: 3D Finite element modeling of dynamic rupture and aseismic slip over earthquake cycles on geometrically complex faults. *Bulletin of Seismological Society of America*, 110(6), 2619-2637, 2020.

2015-2019

- [5] Zhong, S., Wan, Z., Duan, B., **Liu, D.**, Luo, B.: Do earthquakes trigger mud volcanoes? A case study from the Southern margin of the Junggar Basin, NW China. *Geological J.*, 54(3), 1223-1237, 2019.
- [4] **Liu, D.**, Duan, B.: Scenario earthquake and ground motion simulations in North China: effects of heterogeneous fault stress and 3D basin structure. *Bulletin of Seismological Society of America*, 108(4), 2148-2169, 2018.
- [3] Harris, RA, Barall, M., Aagaard, B., Ma, S., Roten, D., Olsen, K., Duan, B, **Liu, D.**, et al.: A suite of exercises for verifying dynamic earthquake rupture codes. *Seis. Res. Lett.*, 89(3), 1146-1162, 2018.
- [2] Duan, B., **Liu, D.**, Yin, A: Seismic shaking in the North China Basin expected from ruptures of a possible seismic gap. *Geophys. Res. Lett.*, 44(10), 4855-4862, <https://doi.org/10.1002/2017GL072638>, 2017.
- [1] **Liu, D.**, Hu, C., Cai, Y.: Numerical simulation of the dynamic rupture process of the 2011 Tohoku-Oki, Japan, Mw9.0 earthquake. *Chinese J. of Geophys.*, 58(9), 3133-3143, 2015 (in Chinese).

PREPRINTS

- [P4] **Liu, D.**, You, K., Ren, P., Tang, H.: Nonlinearly coupled permafrost dynamics from graph neural networks. *Research Square*, <https://doi.org/10.21203/rs.3.rs-10910110/v1>, 2026.
- [P3] Song, C., Puel, S., **Liu, D.**, Wallace, L., Becker, TW.: The role of elastic heterogeneity for inferring seismic-cycle deformation at the Nicoya Peninsula, Costa Rica. *ESS Open Archive*, <https://doi.org/10.22541/essoar.15005894/v1>, 2026.
- [P2] Yun, J., Fialko, Y., May, D., Gabriel, A.-A., Williams, C., **Liu, D.**: Effects of loading schemes in volumetric simulations of sequences of earthquakes and aseismic slip (SEAS) in subduction zones. *EarthArXiv*, <https://doi.org/10.31223/X5D20T>, 2026.
- [P1] Domeyko, R., Partin, J., Okumura, Y., Thirumalai, K., DiNezio, P., et al. (incl. **Liu, D.**): Rare deglacial-aged corals point to weaker El Niño in a colder world. *EarthArXiv*, <https://doi.org/10.31223/X5R50Z>, 2026.

Grants

- 2025-2028 **Forced Changes in Tropical Pacific Variability from Glacial Times to the Future, Co-PI,**
National Science Foundation, Award
- 2023-2026 **Investigation of climate and weather extremes in a super large ensemble of CAM4
simulations, Co-PI, Verisk**
- 2024-2026 **Investigating Periodicities in Abyssal Hill Morphology of the Atlantic Ocean: Possible
Evidence of Mantle Dynamics, Co-PI, National Science Foundation, Award 2341367**
- 2022-2025 **Collaborative Research: Elements: Monitoring Earth Surface Deformation with the Next
Generation of InSAR Satellites: GMTSAR, took over PI after the grant was funded in 2023,**
National Science Foundation, Award 2209807
- 2024 **Generation of Near Source Broadband Ground Motion from Physics-based Dynamic
Rupture Simulations: Continuing A Group Modeling Approach towards better
Characterizing Seismic Hazard for Engineering Applications, Co-PI, Southern California
Earthquake Center, Award 24103**
- 2023 **Generation of Broadband Ground Motion from Dynamic Rupture Simulations: A Group
Modeling Approach towards better Characterizing Seismic Hazard for Engineering
Applications, Co-PI, Southern California Earthquake Center, Award 23121**
- 2022 **Evolution of ground motion characteristics over earthquake cycle timescales, PI,**
Southern California Earthquake Center, Award 22035
- 2022 **Generation of Broadband Ground Motion from Dynamic Rupture Simulations: A Group
Modeling Approach towards better Characterizing Seismic Hazard for Engineering
Applications, Co-PI, Southern California Earthquake Center, Award 22043**
- 2021 **Generation of Broadband Ground Motion from Dynamic Rupture Simulations: A Group
Modeling Approach towards better Characterizing Seismic Hazard for Engineering
Applications, Co-PI, Southern California Earthquake Center, Award 21010**